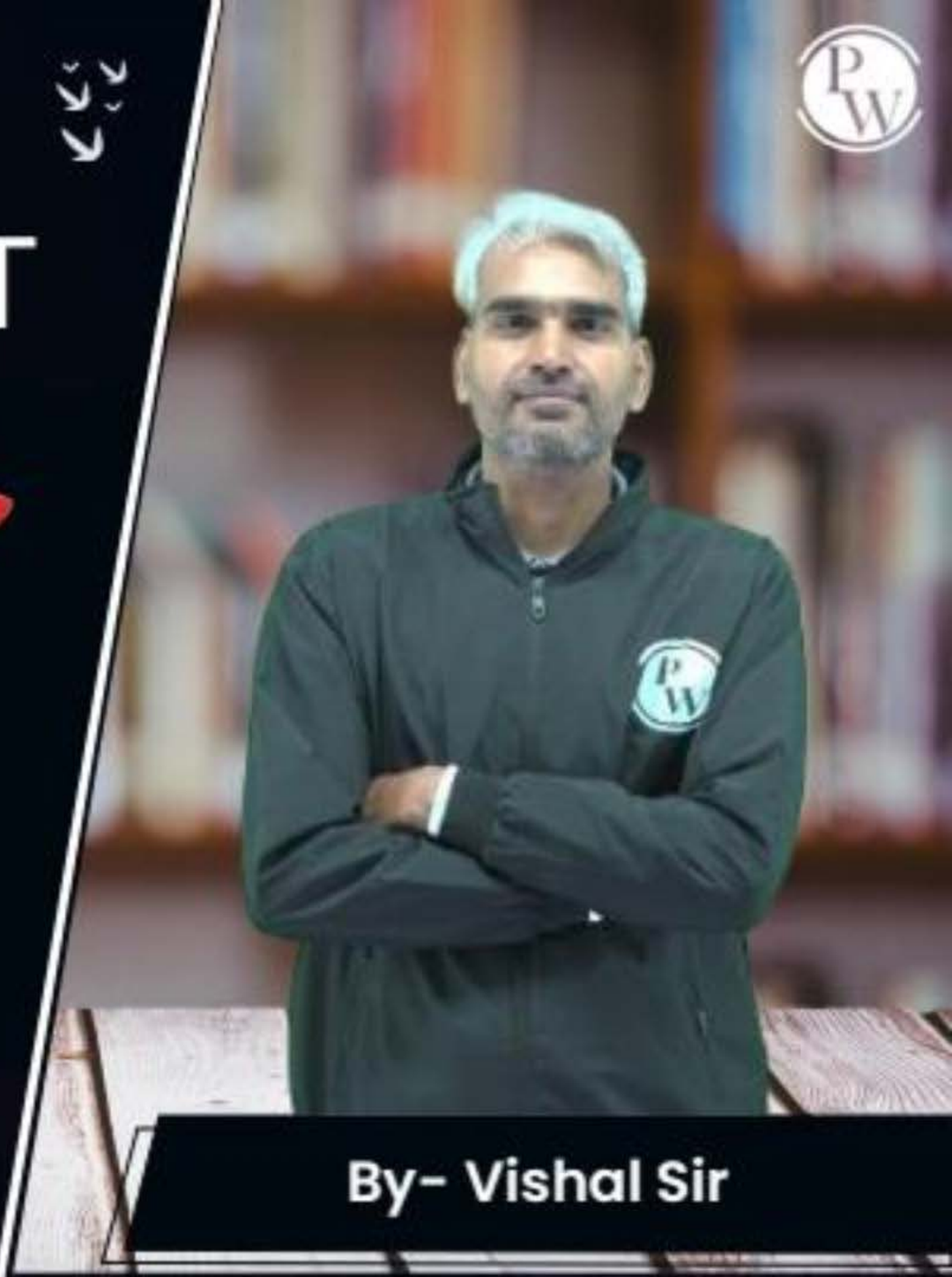
Computer Science & IT



Database Management System

Transactions and Concurrency control

Lecture No. 01



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Recap of Previous Lecture



Topic Domain relational calculus (DRC)

Slide

Topics to be Covered



- **Topic** Transaction and Schedule
- **Topic** ACID properties
- **Topic** Recovery management component

Slide



Topic : Transaction

A sequence of logically related operations which are used to perform a particular task is called transaction.

dataitem

R(A) ≡ Read (A) : It is used to access the value of dataitem A from the database (disk) to main memory

W(A) ≡ Write (A) : Updation of dataitem A into the database (disk)

Transaction to transfer 500 Rs from account A to account B

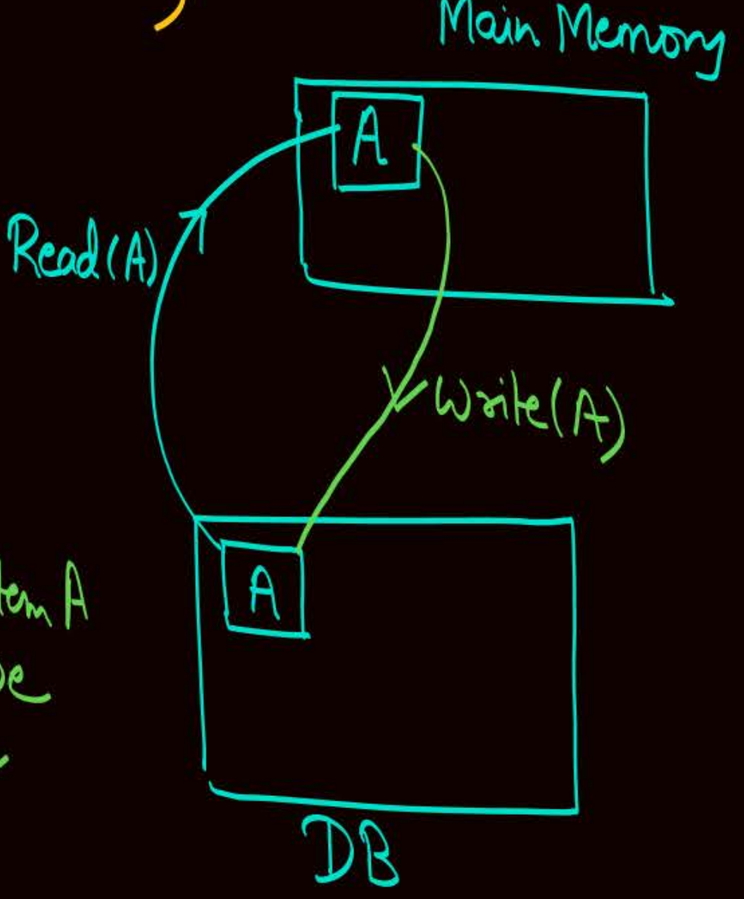
{ Let initial amount in account A = 1000
initial amount in account B = 0 }

Transaction

```
R(A);
A = A-500;  // This updation will be performed only in M.M!

W(A);  // Current value of data item A Present in the M.M. will be updated in the database

R(B);
B=B+500;
W(B);
```



Example of blind write opn :-

Transaction

(i) R(A) ;

(ii) A = A+10 ; // Will be performed only in the main memory

(iii) W(A) ; // New value of dataitem A is updated in DB.

(iv) W(B) ;

Not a blind write opn bcoz of (ii)

W(B) is a blind write opn

If a transaction update (write) any dataitem into database without reading that dataitem from the database then it is called blind write operation.

(v) Commit ; // Transaction Committed means transaction executed successfully {Imp: from user perspective}



Topic : ACID properties

For the integrity of the database transactions must preserve ACID properties

- Atomicity
- Consistency
- Isolation
- Durability



Topic : Atomicity

→ Atomicity states that either execute all the operations of the transaction or execute none of them

Transaction to transfer 500 Rs from account A to account B

{ Let initial amount in account A = 1000
initial amount in account B = 0 }

Transaction

```
R(A);
A = A-500;

W(A);

R(B);
B=B+500;
W(B);
Commit
```

* If transaction fails at this point then integrity of the database may be Compromised

If transaction fails any where in between then we say that atomicity is violated

Reasons for failure of transaction may be
- Power Failure
- S/W Crash
- H/W Crash
- Concurrency Control Components or operating system may kill transaction
- etc

Note: To ensure atomicity if transaction fails before the Commit operation then we must undo the operations performed by that transaction until the failure point

Note:- To ensure atomicity "recovery management Component" is used.

→ Recovery management Component :-

for failed transaction

for committed transaction

- Recovery management component is responsible for "undoing" and "redoing" the operations

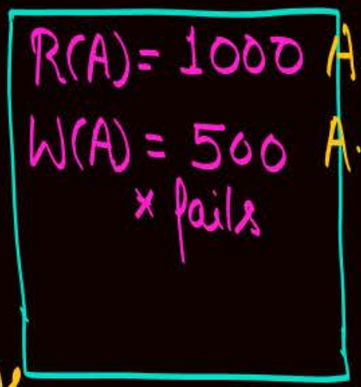
  - The process of undoing the operation performed by a transaction is called "rollbacking" of that transaction

  - For the purpose of undoing and redoing the operations the recovery management component uses "transaction log".

Transaction log:

Transaction log record the activities performed by transactions

Transaction Log wrt transaction to transfer 500 RS from A to B



If transaction fails then we we can update the values of the dataitems in the database by the old values of those dataitems

- → Transaction log is managed by recovery management component
- → We can recover from the failed transaction only if transaction log is available
- → Transaction log will be stored in the secondary Memory
- → Either we can use an unified transaction log for all ongoing transactions or we can use individual transaction log for each transaction



Topic : Durability

* Durability says that we must be able to recover under most of the cases.
* For the purpose of recovery transaction log is required
  - ∴ objective of durability is to preserve transaction log.
* Transaction logs are stored in the secondary memory, ∴ to preserve the transaction logs we must make the secondary memory architecture more robust.

* RAID architecture are defined to make the disk architecture more robust.

RAID ≡ Redundant array of independent disk

- Generally it is used to maintain duplicate copies of disk along with some other qualities

- RAID-0
- RAID-1
- RAID-2
- RAID-3
- etc.

* If transaction fails then Atomicity & Durability Comes into the picture.



2 mins Summary

- **Topic** Transaction
- **Topic** Serial schedule & concurrent schedule
- **Topic** ACID properties
- **Topic** Recovery management component

THANK - YOU